

Skills Summary

Diagnosing Reversed and Non-Equivalent Ratios

Use this reference while completing your worksheet or when studying.

Catching a Reversed Ratio

Rule: The quantity named **first** goes first. “The ratio of A to B” means A’s number, then B’s number. In a rate, the word **per** points at the bottom: miles *per* hour means miles on top, hours on the bottom.

Example: A basket holds 11 lemons and 6 limes. Write the ratio of lemons to limes.

Step 1. Lemons are named first, so the lemon count goes on top of the fraction.

Step 2. 11 lemons to 6 limes gives $\frac{11}{6}$ — flipping it to $\frac{6}{11}$ would answer a different question.

Try it: A drawer holds 9 spoons and 4 forks. Write the ratio of spoons to forks as a fraction.

check: $\frac{4}{9}$

(!) Watch out: A reversed ratio often *looks* fine because it uses the same two numbers. Read the question again and check which quantity was named first.

Testing Whether Two Ratios Are Equivalent

Rule: Simplify both ratios to simplest form. Equivalent ratios simplify to the **same** ratio. Multiplying both parts by the same number keeps a ratio equivalent; **adding** the same number to both parts does not.

Example: Are 8:12 and 10:15 equivalent? Write = or $\neq$.

Step 1. Simplify both: dividing 8 and 12 by 4 gives two-thirds, and dividing 10 and 15 by 5 also gives two-thirds.

Step 2. Same simplest form, so $\frac{8}{12} = \frac{10}{15}$.

Try it: Are 6:9 and 8:12 equivalent? Write = or $\neq$.

check: $\frac{8}{12} = \frac{6}{9}$

Fixing a Broken Ratio Table

Rule: Test every column the same way: divide one row by the other (or simplify the column's ratio) and check it matches the table's ratio. To repair a bad entry, scale the **base column** by the right multiplier.

Example: A 3:5 table pairs a top entry of 12 with a bottom entry of 21. Fix the bottom entry.

Step 1. Find the multiplier from the base column: $12 \div 3 = 4$, so every entry in that column is the base times 4.

Step 2. Bottom entry: $5 \times 4 = 20$, so the corrected entry is **20** (the 21 came from a skip-counting slip).

Try it: A 4:3 table pairs a top entry of 16 with a smudged bottom entry. What should the bottom entry be?

check: 12

Key Vocabulary

Reversed ratio: the right numbers in the same form
wrong order

Multiplier: what both parts are scaled by

Equivalent ratios: ratios that simplify to the **Rate:** a ratio of two different kinds of units